

Dr Hengfei Wang

POSTDOCTORAL RESEARCH FELLOW

Perception and Activity Understanding Group, EPFL & Idiap, Switzerland

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About Me

I am a Postdoctoral Research Fellow at EPFL and Idiap Research Institute, working with Dr. Jean-Marc Odobez on Human Perception and Activity Understanding. I received my PhD in Computer Science from the University of Birmingham in 2025. Prior to that, I earned both my Bachelor's and Master's degrees in Mechanical Engineering from Tsinghua University in 2017 and 2020, respectively.

My research lies at the intersection of vision-language understanding, gaze and attention modeling, social behavior analysis, and 3D-aware digital human generation. I aim to develop AI systems that can perceive, interpret, and generate human-centric behaviors, including where people look, what they attend to, how they interact, and how realistic digital humans can be controlled. I have published at CVPR, ECCV, ACM MM, AAAI, and BMVC, with multiple oral presentations.

Education

University of Birmingham (UoB)

PH.D. IN COMPUTER SCIENCE

Birmingham, UK

Sep. 2020 - Mar. 2025

Tsinghua University (THU)

M.ENG. IN MECHANICAL ENGINEERING

Beijing, China

Sep. 2017 - Jun. 2020

Tsinghua University (THU)

B.ENG. IN MECHANICAL ENGINEERING

Beijing, China

Sep. 2013 - Jun. 2017

Employment

Postdoctoral Research Fellow

PERCEPTION AND ACTIVITY UNDERSTANDING GROUP

EPFL & Idiap

Sep. 2025 - Present

- **Towards a Unified understanding of Attention and Social Behaviors in the Wild**
- **Keywords:** Gaze Following, Social Behaviors Analysis, Multimodal Large Language Models (MLLMs)
- **Introduction:** This research focuses on developing a unified multi-task learning framework to model human social behavior and attention in diverse, real-world environments. By leveraging Transformer-based scene and person encoders with cross-attention fusion, the project integrated subject-level dynamics (gaze, gestures) with scene-level social signals. The research focused on leveraging Vision-Language Models and graph-based message passing to create robust, transferable representations for human-centric computer vision.
- **Line Manager:** Dr. Jean-Marc Odobez

Postdoctoral Research Fellow

SCHOOL OF COMPUTER SCIENCE

University of Birmingham

Mar. 2025 - Sep. 2025

- **Gaze Estimation in the Wild**
- **Keywords:** Gaze Estimation
- **Introduction:** This research focuses on developing robust gaze estimation models that perform reliably in real-world, unconstrained environments. Unlike controlled lab settings, real-world scenarios involve varying head poses, lighting conditions, occlusions, and diverse demographics, which pose significant challenges for accurate gaze tracking. The project aims to bridge this gap using deep learning, domain adaptation, and data augmentation techniques. I am responsible for designing and implementing novel gaze estimation algorithms, conducting large-scale experiments, and contributing to publications. I also collaborate with other researchers to integrate the models into broader human-computer interaction systems.
- **Line Manager:** Dr. Hyung Jin Chang

Professional Activities

ACADEMIC EVENT

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|-----------|--|-------------------|
| 2026 | Organizer , 7th International Workshop on Eye and Gaze in Computer Vision | CVPR Workshop |
| 2025 | Program Committee , 1st Human-Autonomous Vehicle Interaction Workshop | WACV Workshop |
| 2024 | Webchair , 6th International Workshop on Gaze Estimation and Prediction in the Wild | CVPR Workshop |
| 2023 | Webchair , 5th International Workshop on Gaze Estimation and Prediction in the Wild | CVPR Workshop |
| 2022 | Webchair , 4th International Workshop on Gaze Estimation and Prediction in the Wild | CVPR Workshop |
| 2021-2025 | Maintainer of Biggest Gaze Research Repository , Awesome Gaze Estimation in Github | Github Repository |

TEACHING

GRANT

2018-2020 **Participants [£411,552]**, “In-line Inspection of Urban Gas Pipeline Networks”

Beijing Municipal Science & Technology Commission

Peer-Review Activities

WORKSHOP

International Workshop on Gaze Estimation and Prediction in the Wild (2023/2024)
Human-Autonomous Vehicle Interaction Workshop (2025)

CONFERENCE

CVPR(2021/2022/2023), ECCV(2022/2024), AAAI(2024/2025/2026), ICME2025, NeurIPS(2025/2026), ACM MM2025

JOURNAL

IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)

Honors & Awards

2024	ECCV Oral Paper Award , ECCV2024	MiCo Milano, Italy
2020-2024	China Scholarship Council (CSC) Scholarship , China Scholarship Council [£57,600]	Beijing, China
2023	BMVC Student Funding , BMVC2023	Aberdeen, UK
2023	BMVC Oral Paper Award , BMVC2023	Aberdeen, UK
2019	ICRAE Best Oral Presentation Award , ICRAE2019	Singapore
2018	Outstanding Social Work Scholarship , Tsinghua University	Beijing, China
2017	Finalist of Comprehensive Winter Study Abroad Program , University of Tokyo	Tokyo, Japan
2016	Finalist of International Undergraduate Research Experience Program , University of Notre Dame	Notre Dame, USA
2014	China National Encouragement Scholarship , Chinese Government	Beijing, China
2014	Science and Technology Innovation Scholarship , Tsinghua University	Beijing, China
2013	2nd Prize in Mechanical Innovation Competition , Tsinghua University	Beijing, China

Presentations

2025	Gaze-Controllable Face Generation , Invited Talk @ TAM at Idiap	Birmingham, UK
2024	Generalization of Gaze Estimation , Invited Talk @ 2024 CVPR Digestion Workshop at UoB	Birmingham, UK
2023	High-Fidelity Eye Animatable Neural Radiance Fields for Human Face , BMVC2023	Aberdeen, UK
2023	Image Synthesis for Gaze Estimation , Invited Talk @ 2023 CVPR Digestion Workshop at UoB	Birmingham, UK

Publications

* Co-First Author; † Project Leader

INTERNATIONAL CONFERENCE

2026	Eyes on VLM: Benchmarking Gaze Following and Social Gaze Prediction in Vision Language Models Hengfei Wang, Anshul Gupta, Pierre Vuillecard, Jean-Marc Odobez Submitted to NeurIPS2026	
2026	RTGaze: Real-Time 3D-Aware Gaze Redirection from a Single Image Hengfei Wang, Zhongqun Zhang, Yihua Cheng, Hyung Jin Chang AAAI Conference on Artificial Intelligence (AAAI)	
2025	Roll Your Eyes: Gaze Redirection via Explicit 3D Eyeball Rotation YoungChan Choi*, Hengfei Wang*†, Yihua Cheng, Boeun Kim, Hyung Jin Chang, Younggeun Choi, Sang-Il Choi ACM International Conference on Multimedia (ACM MM), Oral Presentation	

- 2025** **3D Prior Is All You Need: Cross-Task Few-shot 2D Gaze Estimation**
 Yihua Cheng, Hengfei Wang, Zhongqun Zhang, Yang Yue, Boeun Kim, Feng Lu, Hyung Jin Chang
Conference on Computer Vision and Pattern Recognition (CVPR)
- 2024** **TextGaze: Gaze-Controllable Face Generation with Natural Language**
Hengfei Wang, Zhongqun Zhang, Yihua Cheng, Hyung Jin Chang
ACM International Conference on Multimedia (ACM MM)
- 2024** **NL2Contact: Natural Language Guided 3D Hand-Object Contact Modeling with Diffusion Model**
 Zhongqun Zhang, Hengfei Wang, Ziwei Yu, Yihua Cheng, Angela Yao, Hyung Jin Chang
European Conference on Computer Vision (ECCV), Oral Presentation
- 2023** **High-Fidelity Eye Animatable Neural Radiance Fields for Human Face**
Hengfei Wang, Zhongqun Zhang, Yihua Cheng, Hyung Jin Chang
The British Machine Vision Conference (BMVC), Oral Presentation
- 2023** **GazeCaps: Gaze Estimation With Self-Attention-Routed Capsules**
Hengfei Wang, Jun O. Oh, Hyung Jin Chang, Jin Hee Na, Minwoo Tae, Zhongqun Zhang, Sang-Il Choi
Conference on Computer Vision and Pattern Recognition Workshop (CVPRW)
- 2019** **Robotic System with Power Line Communication for In-pipe Inspection of Underground Urban Gas Pipeline**
Hengfei Wang, Zandong Han, Qingxian Ma
International Conference on Robotics and Automation Engineering (ICRAE), Best Oral Presentation
- PATENTS**
- CN Patent** **A Method for Rapidly Probing Pipeline Path Changes**
 Zandong Han, Hengfei Wang, Xiaxi Li, Yuangen Wei
CN110296295B [Granted on 24 Apr 2020]